

Soft Magnetoelastic Microneedle Patch for Rapid Skin Cancer Screening

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Abstract

Skin cancer is among the most prevalent malignancies globally, with incidence rising sharply due to increased ultraviolet exposure and an aging population. Early and accurate detection remains a significant clinical challenge, underscoring the urgent need for accessible, noninvasive diagnostic approaches. Here, we introduce a soft magnetoelastic microneedle patch that rapidly maps skin stiffness with high precision, enabling quantitative assessment of soft tissue mechanics across complex surfaces and spatial scales through self-powered magnetoelastic transduction. Covering a 7.7 cm² sensing area, the patch achieves ultrahigh sensitivity (< 2 kPa), high resolution (32 × 32 pixels, 38 ppi), and a frame rate of 2000 Hz, enabling real-time in vivo tissue mapping to depths of up to 7 mm. Beyond detecting tumor stiffness, the platform supports longitudinal monitoring of tumor progression, adjunctive mechanical information that may inform excision planning, and evaluation of therapeutic response, providing mechanobiological insight into tumor dynamics. Validation in animal models and human subjects demonstrates robust performance and translational potential. This platform offers a transformative path for skin cancer diagnostics and mechanobiology, with strong potential for point-of-care diagnostics in resource-limited settings.

Keywords: tissue mechanics, skin cancer, magnetoelasticity, microneedle patch, cancer screening, mechanomedicine

Introduction

Cancer is a multistage pathological process that progresses from precancerous states to malignant disease, driven by dysregulated cellular functions and uncontrolled proliferation at both the cellular and tissue levels¹. These pathological changes can ultimately lead to patient mortality through immune system compromise, excessive metabolic burden, and organ failure². Importantly, early-stage cancers are often highly responsive to curative treatment^{3,4}, with skin cancers showing particularly favorable outcomes when detected promptly⁵. Early diagnosis is therefore critical for optimizing patient survival, as the 5-year survival rate for stage IV melanoma falls below 20%, compared with more than 98% when identified at earlier stages⁶. While advanced skin cancers can often be recognized even by physicians, accurate and low-cost diagnosis of early-stage skin cancer remains a significant challenge.

Conventional modalities for skin lesion assessment, including visual inspection, dermoscopy, reflectance confocal microscopy and ultrasound-based elastography, provide clinically useful morphological or mechanical information but remain limited by operator dependence, cost, instrumentation requirements or restricted availability⁷. Tissue biomechanical properties, particularly stiffness, have emerged as powerful diagnostic indicators. Malignant transformation is accompanied by extracellular matrix remodeling, elevated compressive stresses on adjacent tissues, and dysregulation of cytoskeletal tension, collectively giving rise to pronounced alterations in Young's modulus (**Fig. 1a**)⁸. However, existing stiffness-measurement techniques, such as atomic force microscopy^{9,10} and micropipette aspiration¹¹, which are inherently invasive, as well as magnetic resonance¹², ultrasound elastography^{13,14}, and optical coherence tomography (OCT), which require complex and stationary imaging systems, are largely restricted to research laboratories or specialized clinical facilities (**Fig. 1b** and **Supplementary Fig. 1**). These technological and logistical limitations hinder widespread deployment and preclude point-of-care use, underscoring the urgent need for rapid, low-cost, and accessible stiffness-based approaches to enable population-scale skin cancer screening.

Herein, we present a portable soft magnetoelastic microneedle patch (MMP), evaluated in animal models, healthy volunteers and ex vivo human skin-cancer specimens, that enables low-cost, point-of-care measurement of tissue stiffness and provides a precise, minimally training-dependent approach for improved solid tumor detection. Leveraging magnetoelastic responses to stiffness changes, the soft MMP uses an intrinsically waterproof, self-powered magnetoelastic transduction layer to quantify tissue mechanics across curved surfaces and multiple scales. With a 7.7 cm² sensing area, ultrahigh sensitivity (< 2 kPa), fine spatial resolution (32 × 32 pixels, 38 ppi), and a 2000 Hz frame rate, the soft MMP profiles tissues in vivo up to 7 mm deep, enabling longitudinal, minimally trained monitoring and offering a transformative approach for early skin cancer screening. Assisted by machine learning, soft MMP is integrated into a wireless, miniaturized bioelectronic system capable of high-resolution, multi-scale stiffness mapping, allowing earlier detection and longitudinal monitoring of skin

lesions. Its accessibility, affordability, and noninvasive operation make the soft MMP a transformative platform for large-scale, at-home early skin cancer screening, with the potential to significantly improve the prognosis and patient outcomes.

Results

Developing a soft magnetoelastic microneedle patch

For rapid and noninvasive skin cancer screening and quantification of tumor stiffness, the soft magnetoelastic microneedle patch exploits magnetoelasticity within a compliant polymer matrix. Nanomagnets embedded in silicone microneedles transduce pressure-induced microdeformations at the microneedle–tissue interface into local magnetic flux density changes¹⁵, arising from nanomagnet rotation and spacing modulation (**Fig. 1c**). This transduction mechanism enables self-powered magnetoelastic transduction for tissue-mechanics mapping. As a result, the pressure-responsive microneedles distinguish healthy and malignant skin through stiffness-dependent deformation: stiffer tumor tissue generates larger magnetic perturbations, whereas softer healthy tissue produces attenuated responses, allowing precise, localized mechanical mapping. Importantly, the magnetoelastic microneedles are mechanically compliant (<100 kPa) and deform upon contact, minimizing tissue damage and preserving noninvasiveness.

To measure the tissue stiffness, each microneedle in the soft MMP is coupled to a Hall sensor array, enabling precise detection of microneedle displacement (**Fig. 1d,e**). Stiffness-induced voltage shifts in the Hall sensor, amplified through a transistor, directly reflect changes in the local magnetic field. Exploiting this mechanism, the magnetoelastic microneedles are integrated with a soft, flexible readout circuit to form a fully integrated, all-in-one bioelectronic patch for rapid skin cancer screening (**Fig. 1f,g**). The soft MMP bioelectronic system includes 256 such microneedles (2.78×2.78 cm, 3 mm in height), a flexible circuit board (500 μ m thick), electronic component, and biocompatible covalent polyurethane elastomers to provide measurement accuracy and stability against sweat. When laminated onto the skin, the soft MMP covers a large detection area (7.7 cm²) with high spatiotemporal resolution (pixel density $\sim$ 38 ppi, framerate >2000 Hz) and ultrahigh sensitivity (< 2 kPa), which ensures precise data collection across varying skin contours. Gentle lamination compresses the patch, and the resulting magnetic variations directly reflect local skin stiffness. These spatially resolved mechanical properties serve as the contrast mechanism for image formation (**Fig. 1h,i**).

To further determine the spatial resolution of the soft MMP for tissue stiffness mapping, two adjacent magnetoelastic microneedles were pressed against different tissue phantoms. Magnetic mappings of the microneedles before and after compression revealed changes in the magnetic field density, with greater changes corresponding to the stiffer tissues. Experimental data confirm the system's responsiveness to a range of axial stresses (0–350 kPa), demonstrating both its sensitivity and accuracy in detecting variations in skin stiffness. Consequently, the measured magnetic field density accurately reflects the mechanical properties of the biological

tissues (Fig. 1h). Consequently, the soft MMP allows rapid, noninvasive assessment of the skin stiffness, distinguishing subcutaneous tumors (5–10 mm deep) from healthy tissues by their characteristic stiffness.

Optimization of the soft magnetoelastic microneedle patch

The soft magnetoelastic microneedle patch comprises an array of compliant microneedles embedded with nanomagnets dispersed throughout elastomeric polymer matrices (**Fig. 2a,b** and **Supplementary Fig. 2**). Micro-CT imaging and confocal microscopy revealed the internal architecture and surface topology of the magnetoelastic microneedles, confirming stable nanomagnet–polymer integration and well-defined geometries (**Fig. 2c,d** and **Supplementary Fig. 3**). To identify the matrix formulation that maximizes mechanical-to-magnetic transduction sensitivity, we evaluated magnetoelastic microneedles fabricated from silicone elastomers of increasing stiffness (00-10, 00-20 and 00-30), each loaded with 50 wt.% nanomagnets. Increased normal pressure applied to the magnetoelastic microneedle arrays produced corresponding magnetic field changes, with the softest formulation, the silicone Ecoflex 00-10, exhibiting the steepest pressure–response slope (**Fig. 2e**). This pronounced responsiveness establishes Ecoflex 00-10 as the optimized matrix for subsequent rigidity-mapping measurements.

In addition to tuning the stiffness of silicone matrix, we investigated how nanomagnet loading density influences the device’s mechanical–magnetic transduction performance. Magnetoelastic microneedles were fabricated from silicone containing 25 wt.%, 50 wt.% or 75 wt.% nanomagnets, balancing the expectation that higher nanomagnet loading enhances magnetic field sensitivity but simultaneously increases the composite stiffness. Among these formulations, the 50 wt.% loading yielded the greatest pressure-response sensitivity, followed by 75 wt.% and 25 wt.% (**Fig. 2f**). A similar trend was observed in the stiffer silicone Ecoflex 00-20 matrices, in which the 50 wt.% condition again produced the optimal response (**Fig. 2g**). Collectively, these results indicate that maximal sensitivity arises from the combination of a low Young’s modulus matrix and a intermediate nanomagnet density.

To assess the performance of the soft MMP under controlled mechanical conditions, we quantified its response on tissue-mimicking phantoms spanning Young’s moduli from 0.14 MPa to 1.77 MPa. Magnetic field changes became markedly distinguishable across phantom stiffness levels once the applied pressure exceeded ~10 kPa, and plateaued at a constant value for all phantoms at pressures above ~20 kPa (**Fig. 2h**). Consistent with these observations, independent tests in which the soft MMP was pressed against materials of differing stiffness demonstrated a monotonic increase in magnetic induction with higher Young’s modulus (**Fig. 2i** and **Supplementary Fig. 4**). These results confirm that the soft MMP reliably encodes substrate stiffness into measurable magnetic signatures over physiologically relevant pressure ranges (**Supplementary Fig. 5**).

We next optimized the geometric parameters of the soft MMP by varying the base thickness and microneedle height. Devices with and without magnetoelastic microneedles were first compared to establish the role of needle-mediated deformation. Under increasing normal pressure, patches with a thin 0.2-mm base rapidly reached a

stable magnetic field output (**Fig. 2j** and **Extended data Fig. 1a–d**), whereas those with a thicker 2-mm base displayed larger magnetic field fluctuations and failed to stabilize even at higher pressures (**Extended data Fig. 1e–l**). These results indicate that the magnetoelastic microneedle architecture, and in particular a reduced base thickness, enhances the fidelity of pressure-induction transduction. Consistently, systematic alterations of the base thickness confirmed improved measurement accuracy in thinner-base designs (**Fig. 2k**). We further evaluated magnetoelastic microneedles of different heights (7 mm, 8 mm, and 9 mm), finding that shorter microneedles reached magnetic saturation at lower applied stresses (**Fig. 2l**). Collectively, these findings show that both a thinner base structure and reduced magnetoelastic microneedle height contribute to higher sensitivity and more stable magnetic readouts.

Building on the optimized single-microneedle design, we fabricated a multi-needle array to enable spatially resolved stiffness mapping, with magnetic flux density measured using a magnetometer positioned beneath the soft magnetoelastic microneedle patch (**Fig. 2m,n**). As expected, the magnetic flux density decreased most prominently when microneedles interacted with stiffer substrates, reflecting greater mechanical deformation and enhanced magnetoelastic coupling at those sites (**Fig. 2o**). To assess the performance in a physiologically relevant setting, we conducted finite-element simulations of the patch applied to tissues with heterogeneous mechanical properties in a cancer model. The simulations revealed distinct magnetic signatures for normal versus malignant regions, confirming the array's capability to resolve stiffness gradients across tissue types. Quantitative analysis further demonstrated a monotonic correlation between tissue Young's modulus and the magnitude of magnetic flux variation (**Fig. 2p**), further confirming the ability of the soft MMP to transduce local modulus into measurable magnetic outputs. Together, these findings highlight the fabrication, magneto-mechanical coupling characterization, and system-level optimization of the soft magnetoelastic microneedle patch capable of sensitive and spatially resolved rigidity detection (**Supplementary Fig. 6–9**).

Noninvasive rapid screening and tracking of skin cancers

Many tumor tissues, including skin tumors, generally exhibit greater stiffness than adjacent healthy tissues, making elastic modulus a useful biomechanical biomarker for cancer diagnosis¹⁶. To evaluate the *in vivo* performance of the soft MMP, we conducted stiffness measurements in mice subcutaneously inoculated with B16-Luc melanoma cells (**Fig. 3a,b**) and 4T1-Luc breast cancer cells (**Extended data Fig. 2a–d**). In both tumor models, the soft MMP consistently distinguished tumor-bearing regions from surrounding healthy tissue based on their mechanical signatures (**Extended data Fig. 2e,f**). These stiffness-derived tumor localizations closely aligned with bioluminescence imaging, confirming accurate and reliable *in vivo* detection.

Beyond binary tumor identification, longitudinal monitoring over a 21-day period revealed a progressive, approximately threefold increase in tissue stiffness, reflecting tumor growth and microenvironmental remodeling (**Fig. 3c**). Notably, the soft MMP was capable of detecting tumors whose lateral extent exceeded the spacing between adjacent magnetoelastic microneedles, supporting its applicability for early-stage tumor identification

(**Supplementary Note 1**). Even when tumor sites were not visually apparent, the soft MMP provided quantitative mechanical readouts, capturing a steady increase in elastic modulus throughout disease progression (**Extended Data Fig. 2e,f** and **Supplementary Note 2**). Absolute modulus values were calibrated using independent shear stress-strain measurements (**Fig. 3d** and **Extended Data Fig. 2g**).

We further evaluated the translational potential of the soft MMP in human subjects with skin lesions. Skin lesions span a wide pathological spectrum, ranging from benign nevi to malignant tumors. Benign nevi typically exhibit structural homogeneity and often do not require surgical intervention (**Fig. 3e–h**). In contrast, malignant skin lesions are associated with poor prognosis if not detected at an early stage (**Fig. 3i**). Malignant pathological changes exhibited pronounced biomechanical signatures, characterized by increased tissue rigidity relative to healthy skin and benign lesions, which were readily detected by the soft MMP (**Fig. 3j**). During controlled compression cycles, two types of malignant skin cancers—basal cell carcinoma (BCC) and squamous cell carcinoma (SCC), both demonstrated higher magnetic field changes than healthy skin (approximately 100–200 μT ; **Fig. 3k**, **Supplementary Fig. 10**), corresponding to a two- to threefold increase in elastic modulus after calibration (**Fig. 3l**). These quantitative measurements were consistent with reported human skin stiffness values obtained using suction and torsional testing methods^{17–20}. Collectively, these results demonstrate the robustness of the soft MMP for differentiating diverse skin lesions through mechanical phenotyping.

Furthermore, the soft MMP enables real-time monitoring of skin cancer progression during therapy, supporting adaptive feedback and precise treatment. Its performance was validated in a checkpoint-inhibition cancer model. tumors were established for 10 days, after which mice received intravenous administration of an anti-programmed cell death protein 1 (anti-PD-1) monoclonal antibody, a clinically validated immunotherapy known to induce robust tumor suppression^{21,22}. Following treatment, the soft MMP detected a progressive decline in tissue elastic modulus over an 11-day period, consistent with the reduction in tumor burden expected during effective immune checkpoint blockade. (**Extended Data Fig. 2h–j** and **Supplementary Note 3**).

Additionally, to confirm that stiffness measurements reliably reflected tumor growth and regression, we simultaneously quantified effective radiance using bioluminescence imaging (BLI). Across a 15-day observation window, the soft MMP faithfully tracked tumor expansion and its subsequent suppression after PD-1 blockade (**Fig. 3m–p**). Notably, the temporal trends in elastic modulus closely paralleled the changes in bioluminescent flux, demonstrating that the soft MMP provides a reliable mechanical correlate of tumor burden and enables noninvasive monitoring of immunotherapy responses^{23,24}. Together, these results demonstrate that the soft MMP enables sensitive, noninvasive, and temporally resolved assessment of tumor mechanics *in vivo* (**Supplementary Video 1**), and capability of high fidelity single channel tumor mechanics detection.

Integrating the soft MMP with super-resolution algorithms

After establishing the capability of the soft MMP for temporal stiffness monitoring at a single measurement site, we next evaluated its potential for spatial mapping of the soft tissues. In this context, local variations in mechanical

properties were used as the contrast mechanism for image formation. To benchmark performance, we fabricated standard phantoms representing healthy tissue and a skin-tumor model (**Fig. 4a**). The tumor phantom consisted of a cylindrical, high-modulus elastomer embedded within a soft elastomer matrix to mimic the stiffness heterogeneity of biological tissue. When tested on the normal-tissue phantom, the soft MMP produced a uniform, flat response devoid of spatial contrast (**Fig. 4b**). In contrast, the tumor phantom yielded a distinct stiffness map, with dark color regions corresponding to the embedded 300-kPa inclusion, spatially aligned with the shadowed region shown in **Fig. 4a,c**. Additional calibration using phantoms with a continuous gradient in stiffness confirmed that the soft MMP could accurately resolve graded mechanical properties (**Fig. 4d** and **Supplementary Fig. 11**).

We further examined the ability of the soft MMP to resolve tumor depth. In a phantom with a stiffness inclusion positioned 5 mm beneath the surface, both ultrasound imaging and the soft MMP detected the embedded lesion (**Fig. 4e**). Notably, the MMP provided a clearer delineation of tumor boundaries compared with ultrasound. A summary of measurable depths (**Fig. 4f**) shows that the soft MMP provides high imaging contrast for shallow inclusions (≈ 1 mm), with contrast gradually diminishing as depth increases and becoming negligible at ~ 8 mm. Together, these results demonstrate that the soft MMP, calibrated using standard phantoms, enables spatially resolved and depth-dependent mapping of tissue stiffness (**Supplementary Fig. 12** and **Supplementary Fig. 13**), underscoring its potential as a compact, low-cost platform for mechanical imaging.

To improve user accessibility and enable real-time, portable operation, we integrated the stiffness-mapping platform into a fully wireless measurement system. As illustrated in the system-level block diagram (**Fig. 1i**), the platform incorporates a 64-channel soft MMP array as the front-end sensing module. To ensure high-fidelity signal acquisition, the hardware architecture includes temperature compensation, an instrumentation amplifier, and optional resistor–capacitor (RC) filtering for noise reduction. A serial peripheral interface (SPI) supports sampling at 60 Hz. Magnetic field signals from the soft MMP are routed to the signal-processing block, where all 64 channels are sequentially demultiplexed through a shift register, digitized by a high-resolution analog-to-digital converter, processed by an onboard microcontroller, and wirelessly transmitted to a smart device for visualization and analysis. This integrated system enables rapid, high-throughput stiffness mapping in a compact, user-friendly format.

To address signal degradation arising from the limited number of sensing elements, we incorporated supervised deep-learning–based super-resolution to enhance spatial stiffness mapping. Specifically, we implemented a super-resolution convolutional neural network (SRCNN), which has demonstrated superior performance compared with conventional interpolation methods. In our workflow, the stiffness map generated by bicubic interpolation serves as the input to the SRCNN (**Fig. 4g**). The network comprises three convolutional layers with kernel dimensions of $64 \times 1 \times 9 \times 9$, $32 \times 64 \times 5 \times 5$, and $1 \times 32 \times 5 \times 5$, respectively, with appropriate padding applied to preserve spatial dimensions across layers. A sigmoid activation function introduces nonlinearity.

Training was performed using 70% of the dataset, with 10% reserved for validation and 20% for testing. The SRCNN parameters were optimized by minimizing a peak signal-to-noise ratio (PSNR)-based loss function between the network output and ground-truth stiffness map. As shown in **Fig. 4h**, the reconstructed images more closely resemble the ground truth than the direct bicubic interpolation. Additional validation using an elastomer phantom (**Fig. 4i**) further confirmed that SRCNN substantially improves reconstruction fidelity (**Fig. 4j,k**). These results highlight the advantage of integrating super-resolution algorithms to enhance the accuracy and interpretability of stiffness maps generated by the soft MMP system.

Continuous surveillance of skin cancer recurrence

The soft magnetoelastic microneedle patch can be applied not only for rapid skin cancer screening but also for continuous monitoring of tumor recurrence, addressing a persistent challenge in skin cancer management. Skin cancer recurrence often originates from residual tumor cells and a pro-tumorigenic postsurgical microenvironment, with the wound bed particularly susceptible to accelerated regrowth (Supplementary Note 4). Although early recurrence may be detected during immediate postoperative surveillance, the majority cases emerge months to years later, beyond routine follow-up²⁵. The absence of practical, at-home monitoring tools and the risks of prolonged prophylactic therapy underscore the need for continuous, noninvasive monitoring of excision sites.

To assess the capability of the soft MMP for detecting postoperative tumor recurrence, we compared mice bearing fully excised B16-Luc melanoma tumors (**Fig. 5a**) with mice in which tumors were partially resected (**Fig. 5b**), as confirmed by bioluminescence imaging (BLI). The soft MMP quantified tissue stiffness by measuring magnetic total flux (**Fig. 5c**), revealing a significant difference in skin rigidity between complete and incomplete resections. This stiffness contrast closely mirrored the trends observed in BLI measurements (**Fig. 5d–f**), indicating that the soft MMP can reliably evaluate surgical outcomes at the skin surface. Consistent with these findings, elastic modulus analysis revealed an approximately fourfold increase in tissue stiffness in cases of incomplete tumor removal compared with complete excision (**Fig. 5g,h**).

We next evaluated the malignancy of recurrent tumors based on their morphological characteristics, as highly malignant tumors often exhibit pronounced shape anisotropy^{26,27}. Enabled by the super-resolution algorithms developed in this study, tumor boundaries could be accurately delineated from stiffness maps. Three-dimensional tumor morphology was first characterized using positron emission tomography (PET) imaging, which revealed elevated metabolic activity in tumor regions relative to adjacent healthy tissue, facilitating accurate identification of the tumor and its boundaries (**Fig. 5i–k** and **Fig. 5m–o**). Notably, Young's modulus maps generated directly by the soft MMP showed strong spatial agreement with PET-derived tumor distributions (**Fig. 5l,p**). In contrast to PET imaging, which requires bulky instrumentation and administration of radioactive ¹⁸F-FDG, the soft MMP enables direct, label-free mechanical imaging, offering superior safety and convenience for skin stiffness

assessment and further validating its utility for tumor detection and recurrence monitoring (**Supplementary Fig. 14–16**).

Compared with conventional approaches for skin cancer screening^{28–34}(**Supplementary Note 5** and **Supplementary Table 1**), the soft MMP–embedded bioelectronic system offers four major advantages. **First**, it provides high-resolution mechanical imaging over a large area with complex topology. The soft substrate of the microneedle patch and flexible printing circuit board (PCB) enable the patch conforms to skin with high topology variation, such as bumps and sinking caused by skin lesions. Besides, by integrating deep-learning–based reconstruction, the system achieves a spatial resolution of 32×32 pixels across a 7.7-cm^2 field of view, corresponding to ~ 38 ppi (**Supplementary Note 6 and 7**). **Second**, the platform supports an exceptionally high acquisition rate ($>2,000$ Hz). Through an optimized serial peripheral interface (SPI) and onboard data-compression architecture, the system achieves a refresh rate far exceeding that of conventional ultrasound devices (~ 17 Hz). Such temporal resolution may enable real-time *in vivo* assessment of not only Young’s modulus but also of viscoelastic parameters, including storage and loss moduli. **Third**, the soft MMP reduces dependence on invasive biopsies and eliminates the need for exogenous molecular imaging probes by providing direct, label-free mechanical contrast for cancer detection (**Supplementary Table 2**). **Fourth**, the device is self-operable, easily applied, and low-cost. Fabricated from accessible materials via a simple and scalable process, the MMP system is readily integrated into wireless modules and can be augmented with artificial intelligence for enhanced clinical usability. Overall, the soft MMP represents an innovative diagnostic methodology with direct translational relevance for early skin cancer screening and postoperative cancer management (**Supplementary Note 8** and **Supplementary Table 3**).

Discussion

To validate its clinical potential, the soft MMP was evaluated in both animal models and human subjects, establishing its laboratory and clinical relevance for rapid, noninvasive characterization of skin cancers. This platform enabled time-resolved monitoring of tissue mechanics and revealed how spatial stiffness-mapping images (SSMI) correlate with histological findings and mechanical measurements. Importantly, translating *in vivo* findings to human studies has significant clinical implications, as human skin—particularly aged skin—exhibits stiffness values on the order of ~ 10 kPa, comparable to those observed in murine models. After accounting for surface topology, the soft MMP produced robust and reproducible measurements of human skin rigidity. In summary, the soft MMP provides rapid, noninvasive quantification of skin mechanical properties, offering new insight into time-dependent changes in tissue biomechanics and serving as a sensitive indicator for monitoring therapeutic response. Beyond tumor detection, this approach may be broadly applicable to tracking stiffness variations associated with aging, wound healing, and a wide range of pathological and therapeutic processes. Importantly, unlike conventional microneedle sensing devices that require skin puncture or biomarker extraction, the soft MMP detects biomechanical signals through transient surface indentation. This contact-mode design

reduces the likelihood of skin-barrier disruption and supports repeated non-invasive measurements (**Supplementary Video 2**).

Several limitations remain. First, because the rigidity readout is based on indentation, lesion geometry and surface topography can influence the apparent stiffness signal. For example, a protruding lesion may generate an elevated mechanical response even if its intrinsic tissue stiffness is not high. Future studies should incorporate surface-topography correction, such as three-dimensional skin scanning, to separate geometric effects from intrinsic tissue mechanics. Device geometry may also influence this effect; longer microneedle-shaped microindenters may help maintain contact across uneven skin surfaces, but this design consideration requires systematic validation. Second, we do not claim that mechanics measured by the soft MMP are sufficient for independent skin-cancer diagnosis or subtype classification. Rather, the device provides complementary biomechanical information beyond surface appearance, including rigidity heterogeneity, subsurface mechanical contrast and longitudinal stiffness changes. When interpreted together with visual inspection, dermoscopy and histopathology, these features may support a more integrated decision workflow. At the current stage, reliable classification among BCC, SCC and melanoma remains outside the validated scope of this study. Third, the robustness experiments clarify both the promise and the operating limits of the soft MMP. Although the multichannel readout is sensitive to localized stiffness contrast, it remains influenced by contact mechanics and environmental perturbations. In particular, lateral shear, unstable hydration and variable contact conditions can alter the effective microneedle–skin or microneedle–phantom interface. Future device versions should incorporate alignment guides, anti-slip backing, reference magnetic channels, controlled loading or force logging, and calibration routines to reduce operator and environmental dependence.

Methods

Magnetoelastic microneedle fabrication

Magnetoelastic microneedles can be fabricated using two 3D printing methods. The first method is stereolithography (SLA, Anycubic Photon). In this process, cross-sectional layers of photocurable resin are selectively exposed to ultraviolet (UV) light to solidify the desired geometry. Elastic resin was loaded into the resin tank, and printing was performed in an inverted orientation. The metal build platform descended incrementally into the resin, enabling sequential UV curing of each layer. Standard printing parameters included a 2-s layer exposure time and a 40-s bottom-layer exposure time. To incorporate nanomagnets, 5- μm NdFeB particles were mixed into the resin at 10 wt.% or 30 wt.% until fully dispersed. Increasing nanomagnet loading required proportionally longer UV exposure time; exposure was increased to 4, 6, and 8 s for 10%, 20%, and 30% nanomagnet loading, respectively, with corresponding bottom exposures of 80, 100, and 200 s to ensure complete polymerization. After printing, excess resin was removed using isopropyl alcohol (IPA, Sigma-Aldrich). The magnetoelastic microneedles were then transferred to an ethanol bath and washed for 1 h using an Anycubic Wash and Cure station, removed from the support structures, and post-cured under UV illumination for 1 h. Finally, the printed microneedle arrays were magnetized to full saturation using an impulse magnetizer. Additional fabrication-route details are provided in **Supplementary Note 9** and **Supplementary Fig. 22**.

A second fabrication method employed 3D-printed molds to shape microneedles from soft elastomers. Ecoflex 00-10, 00-20, or 00-30 precursors (parts A and B mixed 1:1) were combined with 25 μm NdFeB microparticles (MQFP-B, Magquench) at 25 wt.%, 50 wt.%, or 75 wt.% and mixed thoroughly. To remove trapped air, the composite was degassed in a vacuum chamber and subsequently poured into the microneedle molds. The

filled molds were cured in an oven at 60 °C for 4 h. After crosslinking, the magnetoelastic microneedle patch was released using acetone to facilitate demolding and then magnetized to saturation using a pulsed magnetizer (0.7–3.5 T). All NdFeB microparticles were coated with a SiO₂ shell via hydrolysis of tetraethyl orthosilicate (TEOS, Sigma-Aldrich) prior to incorporation into the elastomer. Device sensitivity and magnetic response were evaluated using a custom PCB containing a 64-channel Hall sensor array interfaced with an Arduino microcontroller. Magnetic-induction data from each channel were collected and visualized using Processing software.

Magnetic-field readout was performed using a custom PCB integrating an 8 × 8 analog Hall sensor array (64 spatial sensing pixels). The outputs of the Hall sensors were sequentially selected by the onboard multiplexing circuitry and digitized using an onboard 16-bit successive-approximation analog-to-digital converter. Firmware was developed in the Arduino IDE, and the digitized magnetic-induction data were streamed to a host computer for subsequent processing and visualization.

Phantom fabrication

To reproduce the range of stiffness values encountered in biological tissues, we fabricated three classes of elastomer-based phantoms. First, polydimethylsiloxane (PDMS) phantoms were prepared using different crosslinker-to-base ratios (1:35, 1:50, 1:60, and 1:70) to achieve tunable elastic moduli. Second, additional elastomeric phantoms were generated using Ecoflex 00-30, 00-20, and 00-10 (Smooth-On Inc.), which span a lower modulus regime suitable for soft-tissue simulation. Third, alginate hydrogels were produced at concentrations of 3–10% (w/v) to mimic the mechanical properties of very soft tissues. To construct a tumor phantom, a cylindrical insert of stiff elastomer (PDMS mixed at a 1:20 crosslinker-to-base ratio) was embedded centrally within a softer PDMS matrix (1:40 ratio), thereby creating a model with a well-defined stiffness contrast representative of tumor versus surrounding tissue.

Cell lines and animal models

B16-Luc and 4T1-Luc cells were purchased from the American Type Culture Collection (ATCC). Cells were maintained in RPMI-1640 medium supplemented with 10% fetal bovine serum (FBS), 5% penicillin-streptomycin, and 10 µg/mL blasticidin at 37 °C in a humidified incubator with 5% CO₂. All animal procedures were approved by the UCLA Institutional Animal Care and Use Committee (IACUC protocol #2018-083 and #2014-012). Female C57BL/6 and BALB/c mice (8 weeks old, Jackson Laboratory) were used for the melanoma (B16-Luc) and breast cancer (4T1-Luc) models, respectively.

***In vivo* assessment of tumor growth**

B16-Luc and 4T1-Luc tumors were established by subcutaneously injecting 5×10^5 cells in 30 µL PBS into the dorsal skin of C57BL/6 and BALB/c mice, respectively. Tumor progression was monitored by measuring tumor length and width with a Vernier caliper, and tumor volume was calculated as: volume = length × width² × 0.5. Animals were euthanized when tumors reached ~1000 mm³. For the immunotherapy cohort, once 4T1-Luc tumors reached ~40 mm³, anti-PD-1 monoclonal antibody (InvivoGen, 1 mg/kg) was administered intravenously on days 0, 2, and 4 (three total doses).

To establish the incomplete resection model, B16-Luc tumor-bearing mice were anesthetized when tumors reached ~50 mm³. A ~2 cm incision was made along the left and lower margins of the tumor, after which ~90% of the tumor was excised, leaving ~10% in the wound bed; the incision was then closed using Reflex wound clips. Complete tumor removal was performed in a separate cohort as a control. Bioluminescence imaging (BLI) was conducted immediately after surgery to verify resection status. For BLI, D-luciferin (150 mg/kg) was administered intraperitoneally, and images were acquired using an *in vivo* imaging system (IVIS) Spectrum system and analyzed with Living Image software.

***In vivo* assessment of skin tumor size and bioactivity**

Two weeks after tumor implantation, C57BL/6 mice were warmed on a 37 °C heating pad for 30 min and anesthetized with 1.5% vaporized isoflurane. Tumor metabolic activity was assessed using ¹⁸F-fluorodeoxyglucose (¹⁸F-FDG) micro-positron emission tomography/computed tomography (microPET/CT).

Mice received 100 μCi of pharmaceutical-grade ^{18}F -FDG (PetNet Solutions) via tail vein injection, followed by a 1-hour uptake period under light anesthesia. Imaging was performed on a GNEXT combined PET/CT system (Sofie Biosciences). MicroPET data were acquired using a 350–650 keV energy window and reconstructed using 3D-OSEM/MAP algorithms. PET data were decay-corrected, and attenuation correction was applied using the CT images. Co-registered PET/CT volumes were analyzed and quantified using AMIDE and ORS Dragonfly software, with tumors rendered in 3D using Dragonfly.

To further evaluate in vivo tumor size, magnetic resonance imaging (MRI) was also performed. B16 tumor-bearing mice were anesthetized with 1.5% isoflurane and injected with 100 μL of the contrast agent Gd-EOB-DTPA via tail vein. After a 1-hour conscious uptake period, MRI scans were acquired using an Aspect M2 system (Aspect Imaging) with a 35-mm mouse body coil. Following scout imaging, T1-weighted images were collected using TR/TE = 400/8.7 ms, flip angle = 90° , 15 slices at 1 mm thickness, and a 39 mm field of view. Tumors were visualized and analyzed using AMIDE software.

SRCNN for skin rigidity mapping interpolation

Hall sensor matrix is collected at several strain levels from the soft magnetoelastic microneedle patch (20%, 40%, 60%, 80%, 100%). The magnetic induction data is calculated by the absolute difference map from the undeformed status ($\Delta B_z = |B_z(\text{deformed}) - B_z(\text{undeformed})|$). The magnetic induction matrix is then bicubically upsampled to 64 x 64 and used as the SRCNN input.

The model is a classical 3-layer SRCNN with the following architecture: Conv2d (1, 64, 9x9, pad = 4), Sigmoid, Conv2d (64, 32, 5x5, pad = 2), Sigmoid, Conv2d (32, 1, 5x5, pad = 2), Sigmoid. The training loss is the negative peak signal-to-noise ratio (PSNR). Training samples are shuffled each epoch, whereas validation and test samples are processed without shuffling and are evaluated one lesion at a time. Training was performed using 70% of the dataset, with 10% reserved for validation and 20% for testing.

To avoid overfitting, all normalization statistics are computed only from the training subset and then reused unchanged for validation and test data. And model selection is carried out on the validation set, while the test set is held out until the final evaluation of the best validation checkpoint.

Information for clinical tests

Clinical testing on human skin specimens was conducted under protocols approved by the University of California, Los Angeles Institutional Review Board (IRB-23-1293) and the West Los Angeles VA Medical Center Institutional Review Board (VA IRB#1685733). The soft MMP was firstly tested on healthy volunteers with benign nevi for human biocompatibility and calibration. Then excised human skin specimens containing histopathologically confirmed basal cell carcinoma (BCC) and cutaneous squamous cell carcinoma (SCC) were obtained following routine clinical resection. Immediately after excision, specimens were transferred for ex vivo mechanical assessment, and the soft magnetoelastic microneedle patch (soft MMP) was applied directly to the tissue surface to acquire stiffness-related magnetic readouts. All measurements were performed on de-identified specimens and did not alter clinical decision-making.

Statistics and reproducibility

No statistical methods were used to predetermine sample sizes. All data satisfied the assumptions of the applied statistical tests, including normality and homogeneity of variance. Each experiment was independently replicated at least three times, including scanning electron microscopy (SEM), transmission electron microscopy (TEM), optical microscopy, immunofluorescence staining, and fluorescence imaging. Representative data are shown, and consistent results were observed across replicates.

Data availability

Source data are provided with this paper. Other relevant information in this study is included in the Supplementary Information. Further data are available from the corresponding author upon request.

Code availability

The source code of this article is available via GitHub at <https://github.com/719csy/Soft-MMP-for-rapid-skin-cancer-screening> and are available from the corresponding author upon request.

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Author Contributions

J.C. and S.L. conceived the project and supervised the research. X.Z., S.C., and X.H. designed the experiments, performed data analysis, prepared the figures, and wrote the manuscript. X.Z., X.H., S.C., G.L., Y.S., A.L., I.D., and S.X. conducted the animal studies. X.Z., S.C., G.L., X.F., Y.Z., Y.X., Z.D., W.K., G.C., and Z.G. contributed to device design, fabrication, and characterization. J.C. acquired funding for the project. All authors reviewed the manuscript and approved the final version.

Competing Interests

A patent application related to this work has been filed by UCLA, listing J.C., S.L., X. Z. and S.C. as the inventors. The other authors declare no competing interests.

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Figures and Caption

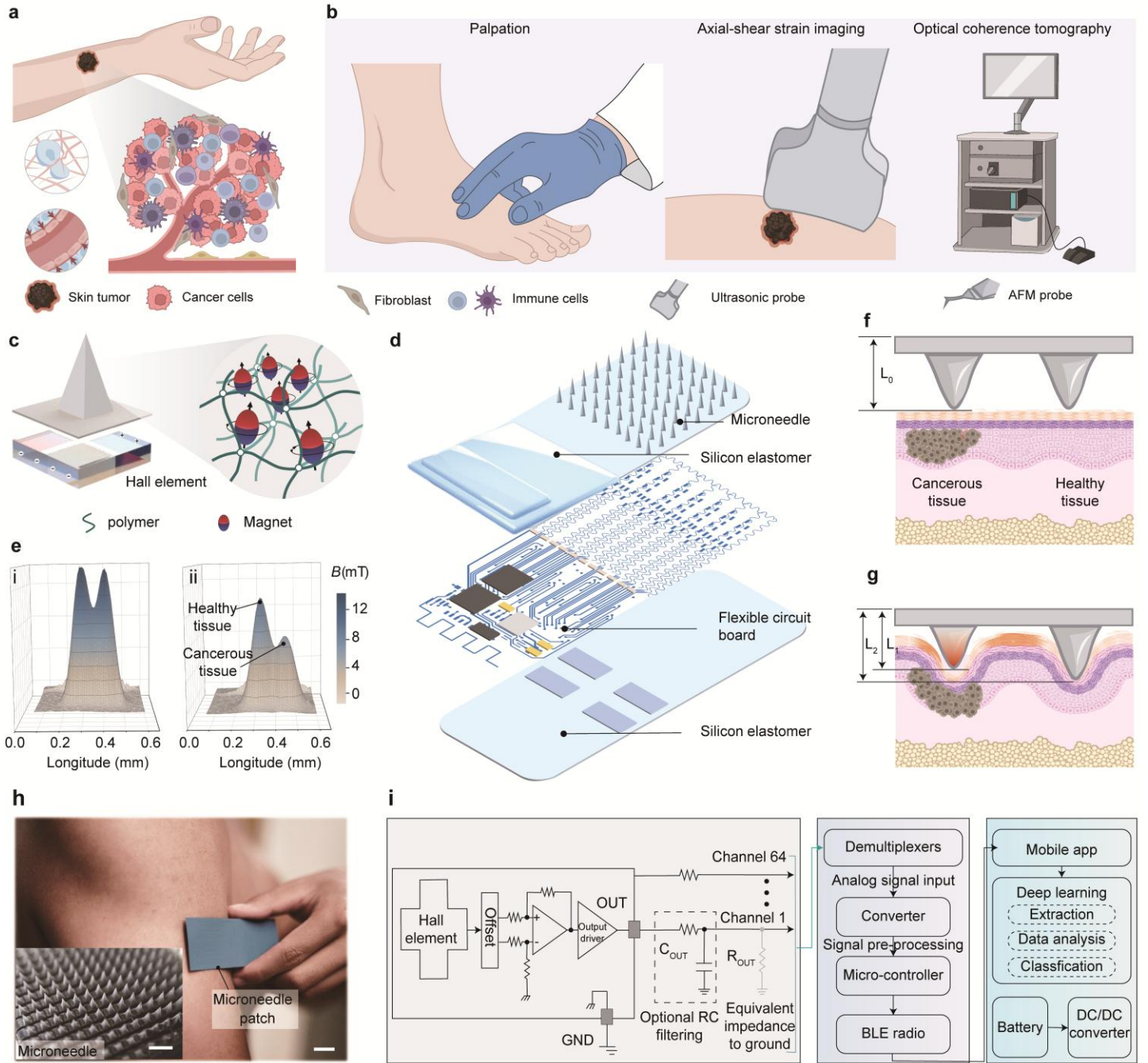


Fig. 1 | An overview of the soft magnetoelastic microneedle patch for rapid skin cancer screening. a, Schematic of skin cancer, its microenvironment, and key contributors to increased tissue rigidity, including extracellular matrix stiffening and elevated interstitial fluid pressure. **b**, Existing approaches for skin cancer detection and rigidity assessment: conventional clinical imaging modalities (for example, ultrasound- and OCT-based methods) require bulky instrumentation and trained operators, whereas atomic force microscopy enables cell- and tissue-level stiffness measurements primarily in preclinical settings. **c**, Schematic of a soft magnetoelastic microneedle integrated with a Hall sensor for magnetic-field-based stiffness sensing. **d**, Schematic of the stretchable electronic readout circuit integrated with the soft MMP. **e**, Magnetic field maps of two microneedles before and after compression. **f, g**, Illustrations of microneedle contact with healthy skin (**f**) and cancerous skin (**g**). L_0 denotes the undeformed microneedle length, whereas L_1 and L_2 indicate the compressed states upon contact with a stiffer and softer substrate, respectively. **h**, Photograph of the microneedle patch applied to the skin. Scale bar, 1.5 cm; inset scale bar, 3 mm. **i**, System-level block diagram of the wireless soft MMP-based wearable platform for skin cancer screening.

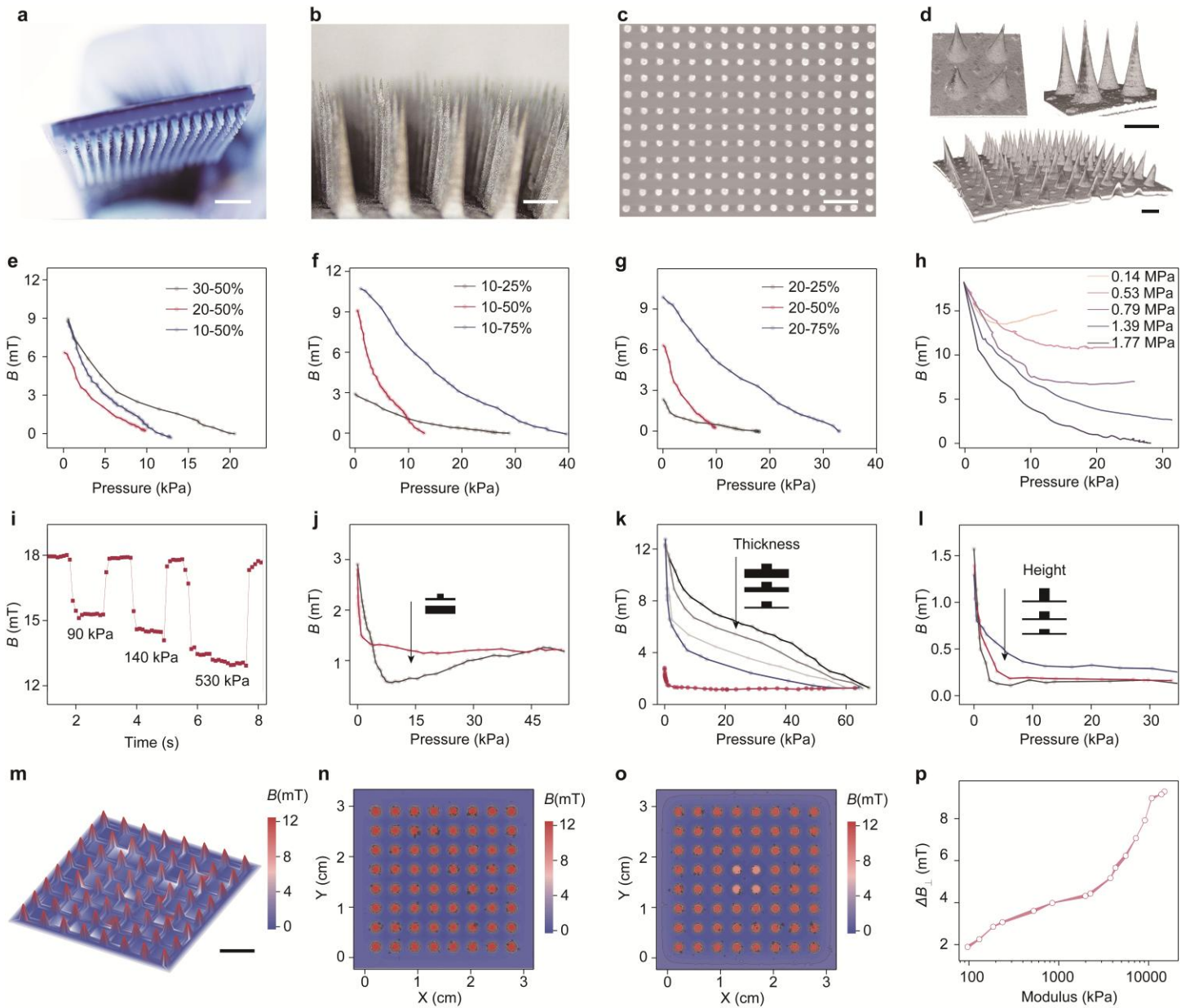


Fig. 2 | Working principle and the mechanical-magnetic coupling investigation of the soft magnetoelastic microneedles. **a**, Photograph of the soft MMP. Scale bar, 5 mm. **b**, Magnified image of individual microneedles. Scale bar, 0.3 mm. **c**, Micro-CT reconstruction of the soft MMP. Scale bar, 3 mm. **d**, Confocal images of the soft MMP. Scale bars, 2 mm. **e**, Applied force versus corresponding magnetic-field changes of the soft MMP. **f**, Magnetic-field response of Ecoflex 00-10 microneedles with different nanomagnet loading density under increasing forces. **g**, Magnetic-field response of Ecoflex 00-20 microneedles with different nanomagnet density under increasing forces. **h**, Magnetic flux-density variation of the MMP on elastomers with different Young's moduli under increasing pressures. **i**, Magnetic flux density variation on elastomers with Young's moduli of 90, 140, and 530 kPa. **j**, Magnetic-field variation as a function of applied force for soft MMPs with different base thicknesses. **k**, Comparison of magnetic-field responses across varying base thicknesses under increasing pressure. **l**, Magnetic field response of microneedles with different heights under applied force. **m**, Magnetic mapping of soft MMP. Scale bar, 2 mm. **n**, Vertical planar magnetic-field distribution of the MMP under no load. **o**, Vertical planar magnetic-field distribution under applied stress, showing greater flux variation at the central four microneedles relative to adjacent ones. **p**, Measured magnetic-flux-density changes on elastomers spanning Young's moduli from 100 to 10,000 kPa.

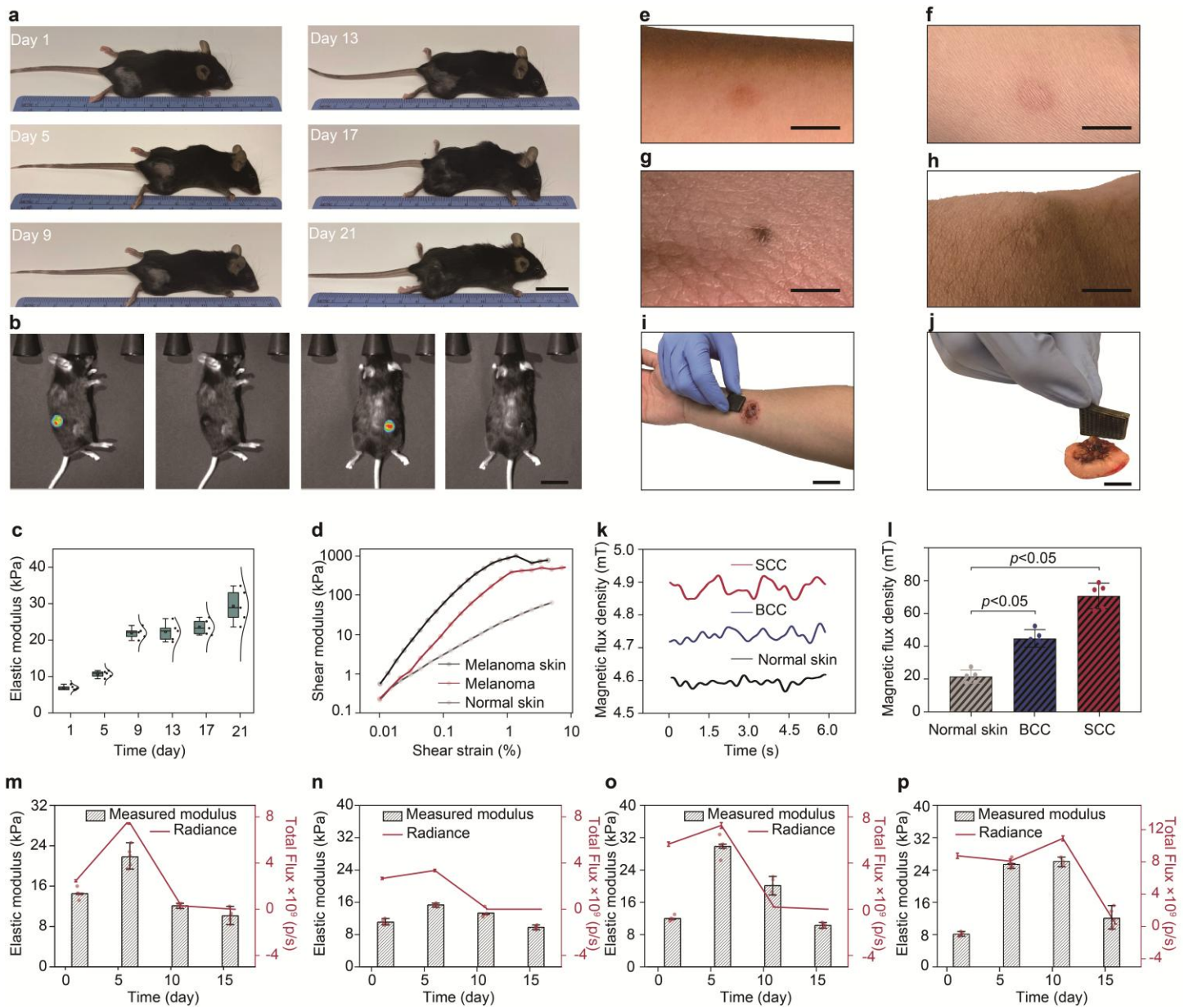


Fig. 3 | *In vivo* spatial mapping of the skin tumor stiffness. **a**, Visual representation of B16-Luc tumor growth over the span of 21 days. Scale bar, 2 cm. **b**, Sagittal (left two) and coronal (right two) images of B16-Luc tumors in mouse models. Scale bar, 2 cm. **c**, Elastic modulus measurements of B16-Luc tumors compared with adjacent healthy tissue using the soft MMP. **d**, Representative shear stress-strain curves of melanoma tumor tissue and healthy tissue, independently used to calibrate the absolute elastic modulus values measured by the soft MMP. **e–h**, Representative clinical photographs of benign human skin lesions (including scar and nevi) and surrounding skin. Scale bars, 1 cm. **i**, Photograph of the soft MMP applied to a human skin lesion for in situ stiffness measurement. Scale bar, 2 cm. **j**, *Ex vivo* stiffness measurement on an excised human skin cancer specimen using the soft MMP. Scale bar, 1.5 cm. **k**, Representative magnetic-flux-density traces recorded during controlled compression cycles on excised normal skin and malignant lesions (basal cell carcinoma, BCC and squamous cell carcinoma, SCC). **l**, Summary of magnetic-flux-density changes measured for normal skin, BCC, and SCC. **m–p**, Comparison of elastic modulus measurements from the soft MMP with bioluminescence radiance across five measurement sessions during tumor growth and subsequent inhibition in four tumor-bearing mice. Data are mean $\pm$ SD; $n = 5$.

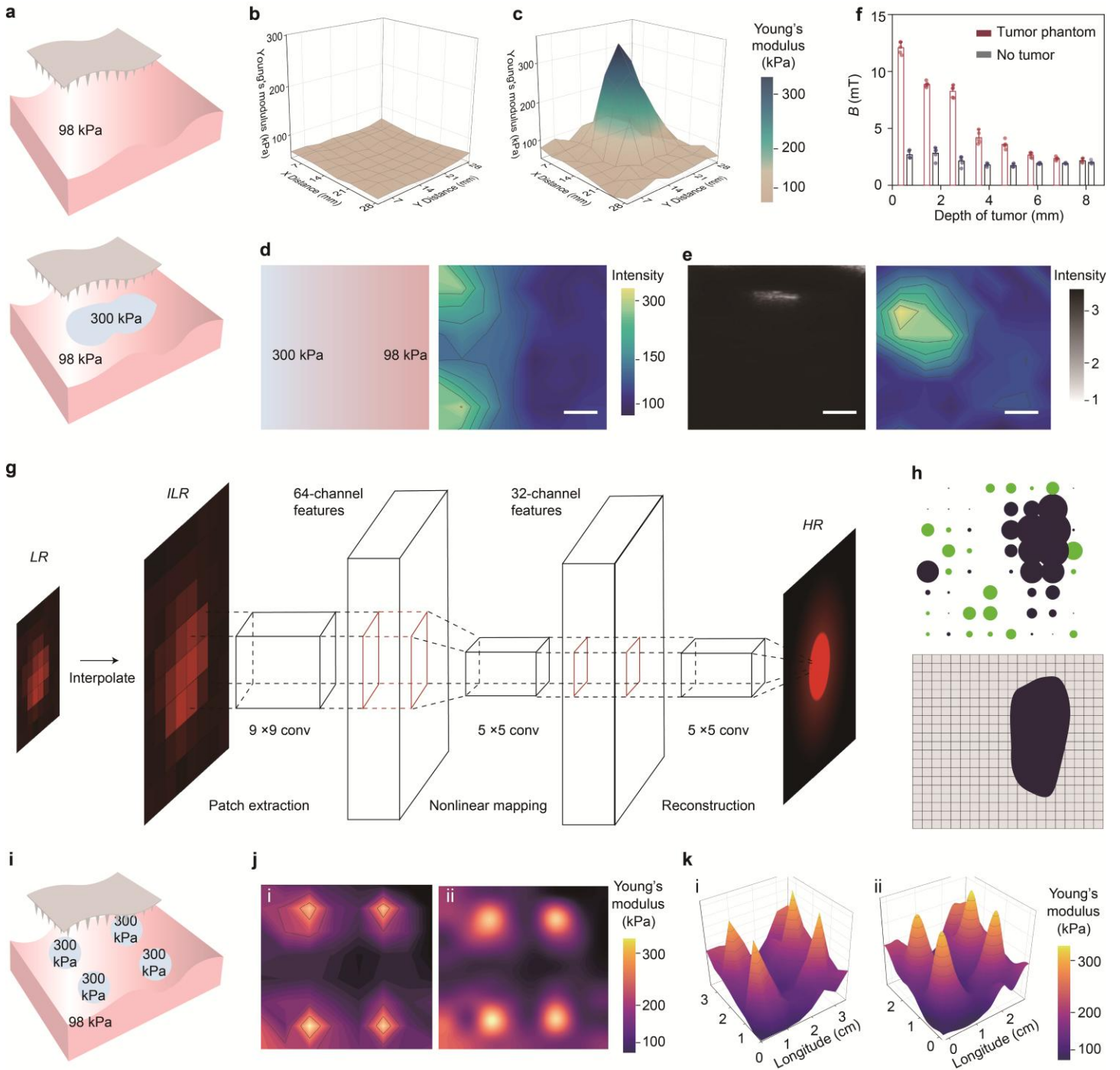


Fig. 4 | Artificial intelligence-assisted soft magnetoelastic microneedle patch for super-resolution mechanical mapping of skin tumors. **a**, Schematic of an elastomer phantom with a uniform modulus of 98 kPa and a central cylindrical inclusion of 300 kPa surrounded by 98-kPa material. **b–c**, Young's modulus maps generated by the soft MMP for the uniform phantom (**b**) and the tumor phantom containing the high-modulus inclusion (**c**). **d**, Spatial stiffness map of a phantom with a continuous modulus gradient from 300 to 98 kPa, reconstructed using the soft MMP. Scale bar, 1 mm. **e**, Comparison of stiffness maps obtained by ultrasound imaging and the soft MMP in a phantom containing a high-modulus region in the upper-left quadrant. Scale bars, 1 mm. **f**, Magnetic-field measurements acquired using the soft MMP for phantoms containing a tumor-mimicking stiff inclusion positioned at depths of 1–8 mm. **g**, Network architecture of the super-resolution convolutional neural network (SRCNN). LR, low resolution, ILR, interpolated low resolution. HR, high resolution. **h**, Original and SRCNN-processed stiffness maps derived from magnetic-field measurements. **i**, Schematic diagram of the phantom used for SRCNN validation. **j**, Original magnetic-field map captured using an 8 × 8 Hall-sensor array and the corresponding bicubic-interpolated map. **k**, Young's modulus maps derived from the original and interpolated magnetic-field data.

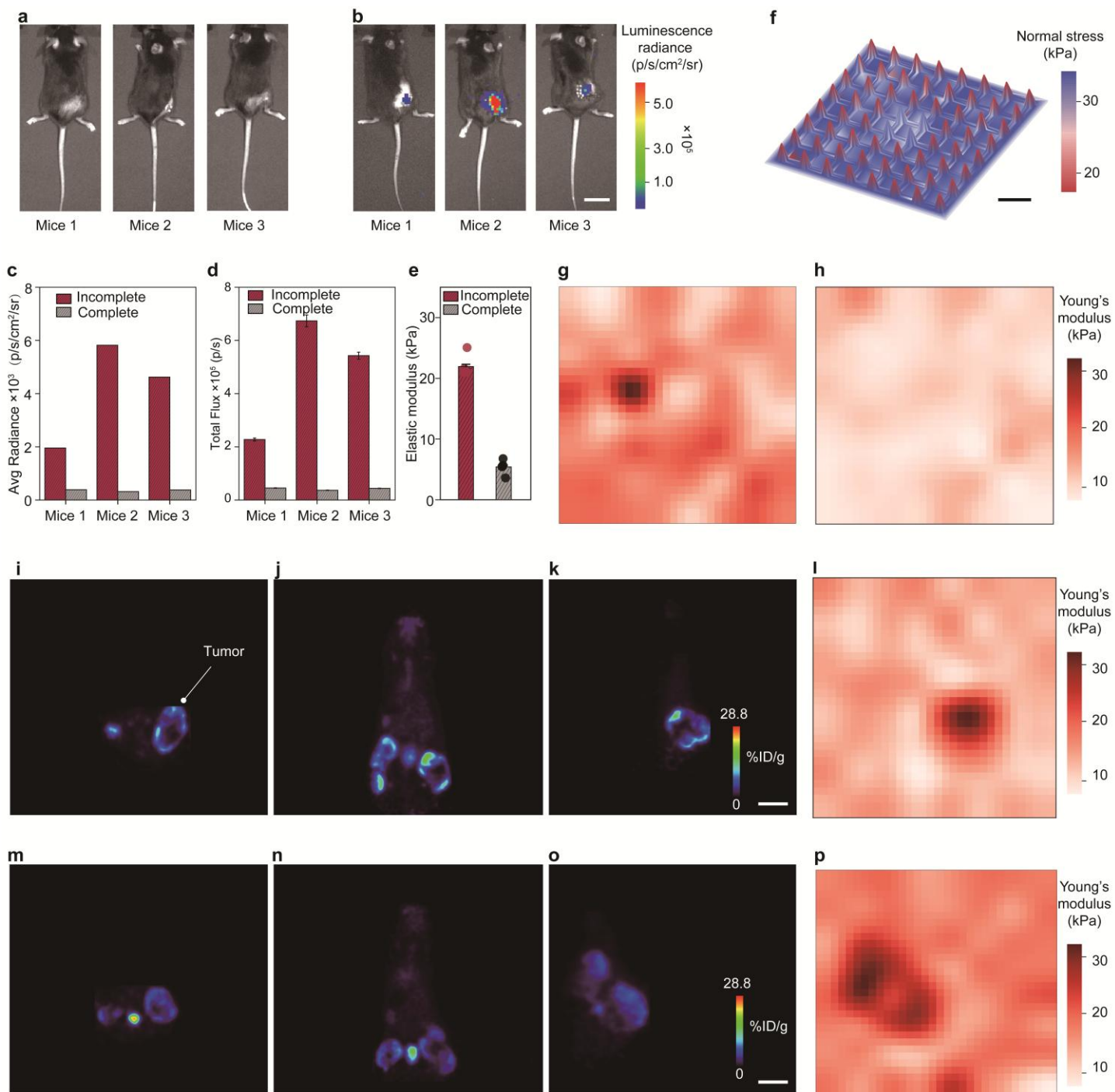
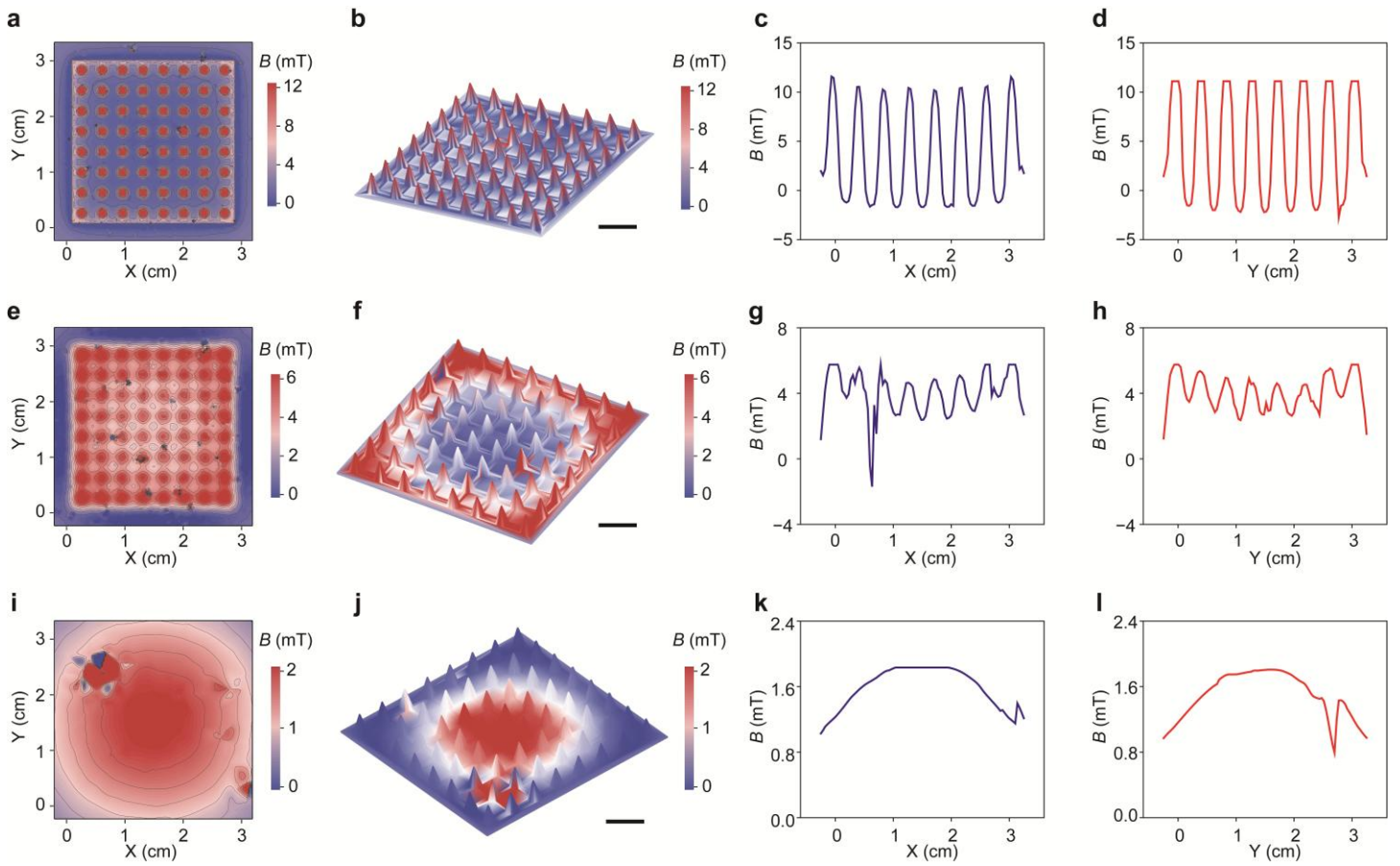
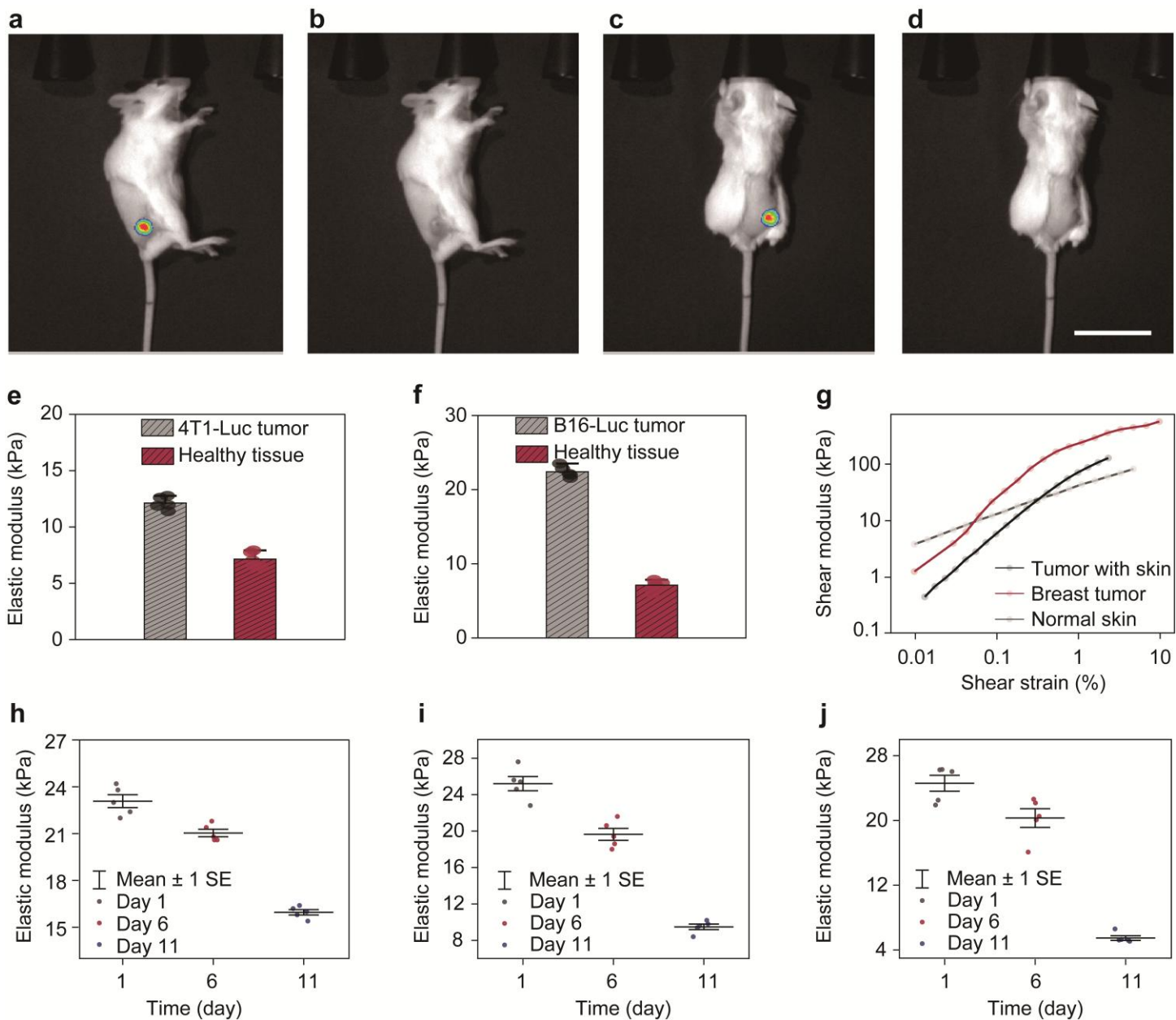


Fig. 5 | *In vivo* tracing of skin cancer recurrence progression. **a,b**, Longitudinal bioluminescence imaging of B16-Luc tumors following complete resection (**a**) and partial resection (**b**) in mice. **c**, Magnetic-field mapping registered with the microneedle patch, showing normal-stress-dependent skin rigidity changes. **d,e**, Average radiance (**d**) and total flux (**e**) measured in mice with complete and incomplete tumor removal. Data are mean $\pm$ SD; $n = 3$. **f**, Elastic modulus values obtained using the soft MMP in mice with complete versus incomplete tumor resection. Data are mean $\pm$ SD; $n = 3$. **g,h**, Young's modulus maps derived from magnetic-flux variations for incompletely removed tumors (**g**) and normal tissue (**h**). **i-k**, MicroPET images of a B16-Luc tumor-bearing mouse in coronal (**i**), axial (**j**), and sagittal (**k**) views, illustrating isotropic tumor morphology. **l**, Isotropic Young's modulus map generated by the soft MMP. **m-o**, MicroPET images of a B16-Luc tumor-bearing mouse in coronal (**m**), axial (**n**), and sagittal (**o**) views, showing anisotropic tumor growth. **p**, Anisotropic Young's modulus map obtained using the soft MMP.



Extended data Fig. 1 | Magnetic induction characterization of the soft magnetoelastic microneedle patch at varying Hall-sensor-to-patch distances. **a,e,i**, Planar magnetic-induction maps acquired at progressively increasing detection distances. **b,f,j**, Corresponding three-dimensional magnetic-field distributions revealing changes in peak definition and field uniformity. Scale bars, 0.5 cm. **c,g,k**, Magnetic-induction line profiles along the x-axis. **d,h,l**, Magnetic-induction line profiles along the y-axis. From top to bottom, increasing the detection distance causes the magnetic-induction patterns to transition from discrete microneedle-resolved signatures to smoother, continuous distributions, highlighting the importance of optimizing sensor–patch spacing for accurate magnetic readout. These observations further indicate that both the Hall-sensor–to-patch distance and the skin–microneedle interface–to-tumor depth can influence the detection efficiency of skin-cancer screening.



Extended data Fig. 2 | *In vivo* stiffness-based identification and longitudinal monitoring of skin tumors using the soft magnetoelastic microneedle patch. **a–d**, Representative bioluminescence imaging of mice bearing subcutaneous 4T1-Luc tumors, showing tumor localization (**a,c**) and corresponding bright-field images (**b,d**). Scale bars, 2 cm. **e,f**, Elastic modulus measured by the soft MMP for 4T1-Luc tumors (**e**) and B16-Luc tumors (**f**) compared with adjacent healthy tissue, demonstrating significantly increased stiffness in tumor regions. Data are mean $\pm$ standard deviation. **g**, Representative shear stress–strain curves of breast tumor, tumor-associated skin, and normal skin, used to independently calibrate absolute elastic modulus values obtained by the soft MMP. **h–j**, Longitudinal elastic modulus measurements during anti-PD-1 immunotherapy, showing a progressive reduction in tissue stiffness over 11 days for 4T1-Luc. Points represent individual measurements; error bars denote mean $\pm$ 1 standard error (SE). $n = 5$.